Part 3: Registers

1. **Name all general-purpose registers in 8086.** AX, BX, CX, DX.
2. **What is the difference between 8-bit and 16-bit registers?** A 16-bit register (like AX) holds 2 bytes. It can be split into two 8-bit registers (AH and AL) that hold 1 byte each.
3. **What is the purpose of the AX register?** It is the **Accumulator**, used for most arithmetic, logic, and data transfer operations.
4. **Explain the function of the BX register.** It is the **Base** register, often used to hold the base address for memory access.
5. **What is the purpose of the CX register?** It is the **Count** register, used automatically as a counter for loops and string operations.
6. **What is the purpose of the DX register?** It is the **Data** register, used for I/O port access and to hold the upper half of large numbers in multiplication/division.
7. **Name the segment registers and their use.** CS (Code), DS (Data), SS (Stack), ES (Extra). They point to the starting addresses of the memory segments where code, data, and stack are stored.
8. **What is the difference between SP and BP?** SP (Stack Pointer) automatically points to the top of the stack. BP (Base Pointer) is used to manually access data *within* the stack (like parameters).
9. **What is the function of SI and DI registers?** SI (Source Index) and DI (Destination Index) are used mainly in string operations to point to source and destination data.
10. **Explain the role of the FLAGS register.** It contains status bits (Zero, Carry, Overflow, etc.) that indicate the result or status of the most recent operation.
11. **Which register is used for string operations?** SI and DI (Source and Destination Indexes) are the primary registers.
12. **How is the IP register used during execution?** IP (Instruction Pointer) holds the offset address of the *next* instruction the CPU needs to execute.
13. **Which registers can be used for pointer arithmetic?** BX, BP, SI, and DI are the index and base registers used for memory addressing.
14. **How is the AX register split into AH and AL?** The top 8 bits are addressed as AH (High), and the bottom 8 bits are addressed as AL (Low).
15. **What is the importance of the CS register?** CS (Code Segment) points to the memory section where the executable code lives. Without it, the CPU cannot fetch instructions.